

Find the slope of the line tangent to the following polar curves at the given points. At the points where the curve intersects the origin (when this occurs), find the equation of the tangent line in polar coordinates.

1. $r = 4 + \sin \theta$; $(4, 0)$ and $(3, \frac{3\pi}{2})$.

2. $r = 8 \sin \theta$; $(4, \frac{5\pi}{6})$.

3. $r = 2 \sin 3\theta$; tips of the leaves.

Find the point(s) at which the following polar curves have a horizontal or vertical tangent line.

4. $r = 3 + 6 \sin \theta$.

5. $r = 4 \cos \theta$.

6. $r = \sin 2\theta$.

Make a sketch of the region and its bounding curves. Find the area of the region.

7. The region inside the leaves of the rose $r = 3 \sin 2\theta$.

8. The region inside the circle $r = 8 \sin \theta$.

9. The region inside the limaçon $r = 2 + \cos \theta$.